Homework Assignment #2

Due by 11:59pm on Friday, October 17, 2025

Purpose and Materials Application
The purpose of this homework assignment is to demonstrate how DFT can be used to explore the electronic and elastic properties of materials. The electronic properties of materials are critical to many different applications, including solar cells. For example, Si is the most commonly used material for solar cells. In this device, a photon excites an electron from the valence band to the conduction band. This excitation process creates a hole in the valence band, and thus an electron-hole pair. An electrical current can be created if the electron and hole can be physically separated. Otherwise, the electron and hole recombine and no current flows. Decreasing or suppressing electron-hole recombination therefore helps to increase the efficiency of the solar cell. We will consider one of the factors that affects the recombination rate in Exercise 1C.

In the second part of the homework assignment you will learn how to calculate the elastic properties of materials using DFT. The elastic properties of materials essentially determine their mechanical behavior but also their thermal properties, as you will learn in the third homework assignment.

Complete the following exercises and answer the relevant questions. **Your assignment write-up should also include the Introduction and Summary/Conclusion sections that you included in the first homework assignment.**

Statement on use of Generative AI: The course is structured such that students should not have to rely on generative AI tools to complete assigned work. You may use generative AI to polish text that you have written, or to debug or write code and you must disclose your use of generative AI. Students are responsible for ensuring that all submitted work is accurate and complete. Copy-pasting text directly from generative AI tools into submitted work is forbidden and will be considered a violation of the Code of Academic Integrity. Since there are no reliable tools to detect generative AI-written text, I expect students to be honest (with me, and themselves) when they submit work. Obvious violations will result in a zero for the assignment and the final course grade will be reduced by half a grade. Students must also ensure that all cited references a) exist, and b) are being properly cited.

Exercise 1: Calculation of the density of states and electronic band structure
We will first calculate the band structure and density of states of Si. Converge the ***lattice constant*** of Si with respect to the plane wave cutoff and **k**-point mesh, as you did in the first homework assignment.

A sequence of calculations are required for determining the electronic band structure and density of states and the various inputs for these calculations can be included in a single submit script. The first calculation in the sequence is simply a total energy (scf) calculation at the relaxed and converged lattice parameter. The **k**-point mesh is much denser than that required for convergence of the lattice parameter. This is because the band structure and DOS are sensitive to the **k**-point mesh density. The second calculation in the sequence is a bands calculation:

```
calculation = 'bands'
```

In this calculation, we calculate the electronic bands along a continuous path through the Brillouin zone. The path is specified in the `K_POINTS` block as follows:

```
K_POINTS crystal_b
3
0.500 0.500 0.500 20
```

```
0.000 0.000 0.000 20
1.000 0.000 0.000 20
```

The keyword following `K_POINTS` specifies that the **k**-points will be listed in crystal coordinates. We have requested that the path pass through three high-symmetry points in the Brillouin zone: $L = (1/2, 1/2, 1/2)2\pi/a$ to $\Gamma = (0, 0, 0)$ to $X = (1, 0, 0)2\pi/a$, with 20 points along each segment. We have also specified the number of bands we would like Espresso to calculate using the `nbnd` keyword, in this case the 4 valence bands and 4 conduction bands of Si. By default, the code will only calculate the occupied bands because including more bands in the calculation increases the computational expense.

The next calculation in the sequence produces a file suitable for plotting the bands (`$PREFIX.bandsplot`), while the remaining two sets of calculations produce the total and partial densities of states. A post-processing program called `plotband.x` can read `$PREFIX.bandsplot` and produce a Postscript file for viewing. First we need to make the program available to us at the command line by loading the Espresso module. In your submit scripts (those with a .sh extension), there is a series of `module load` commands just before `ibrun` execute command. Type those `module load` commands at the command line.

Then execute the program at the command line and enter the name of the relevant file when asked:

```
login3.stampede(1)$ plotband.x
Input file > Si.bandsplot
```

The next few lines deal with the energy range for plotting the bands. You should select a range that runs from just below the lowest energy band to just above the highest energy (here I have selected -8 eV as my lowest energy and 18 eV for the highest energy; note that in contrast to the total energy, which is reported in Ry energy units, the band energies are reported in eV):

```
Reading 8 bands at 21 k-points
Range:  -5.8810 16.3390eV Emin, Emax > -8 18
```

Your energy range may be slightly different. The program detects that the bands have been calculated along a path containing a series of high-symmetry points in the Brillouin zone and prints out the coordinates of the points. This should match what you specified in the `K_POINTS` block for the `'bands'` calculation. The program then asks what it should call the output file that it will produce. It will produce two files, one in xmgr format (this is the format for an old plotting program – Grace – which is no longer maintained or developed) and the other in Postscript format.

```
high-symmetry point:  0.5000 0.5000 0.5000 x coordinate 0.0000
high-symmetry point:  0.0000 0.0000 0.0000 x coordinate 0.8660
high-symmetry point:  1.0000 0.0000 0.0000 x coordinate 1.8660
output file (xmgr) > Si_bands.xmgr
bands in xmgr format written to file Si_bands.xmgr
output file (ps) > Si_bands.ps
```

The program then asks for the Fermi energy. The Fermi energy can be anywhere in the gap for an insulator, so instead we input the energy of the highest occupied electronic state; this is a standard convention in the computational materials literature. In the output file for the scf part of the calculation, you should see a line like this (the actual values of the energies may be slightly different):

```
highest occupied, lowest unoccupied level (ev):  6.2108 6.7023
```

So for the Fermi energy we would input:

```
Efermi > 6.2108
```

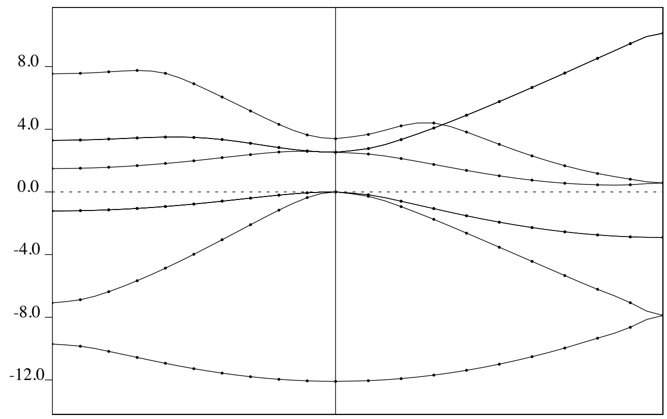

Figure 1: An example of a poorly presented band structure for Si, calculated using the procedure described above. Plots presented without axis labels will result in points being deducted from the final score for the homework assignment.

Now we have to enter some information that tells the program how to plot the bands: the energy increment and the energy to which the other energies are referenced (the 'zero').

```
deltaE, reference E (for tics) 4 6.2108
bands in PostScript format written to file Si_bands.ps
```

Our choice of zero is the energy of the highest occupied electronic state. The program will make this energy 'zero' and shift all the other energies accordingly. Finally, a Postscript file is produced for viewing. It should look something like the plot in Figure 1.

The procedure for plotting the densities of states is more straightforward and will be discussed in class.

1A Calculate the band structure and density of states for Si using the procedure described above. Prepare plots of the total density of states, and the density of states for each kind of orbital (s, p, d). Assign peaks in the total density of states to atoms/orbitals using information from the atom- and orbital-resolved densities of states. Prepare your plots such that the highest occupied energy level is set to zero energy. What is the chemical (orbital) character of the valence and conduction bands in Si? Compare your calculated band structure with results from other DFT calculations in the literature. Hint: it may be easier to first locate the relevant papers in the literature and then prepare your band structure plot using the same path through the Brilloiun zone. Hence, you may need to change the information in the `K_POINTS` block in the `'bands'` part of the calculation.

1B What is the band gap for Si? Is Si a direct or indirect gap insulator? How does your gap compare with experiment, and with other DFT calculations in the literature?

1C Examine the DOS for bulk Si and Si doped with Au in Figure 2. Pay particular attention to the DOS at the band gap. Now read *Nature Materials* **4** 676 (2005). How do you think the presence of Au affects the performance of Si in solar cells?

1D In this exercise we will consider the dielectric material $BaTiO_3$. $BaTiO_3$ is an industrially important

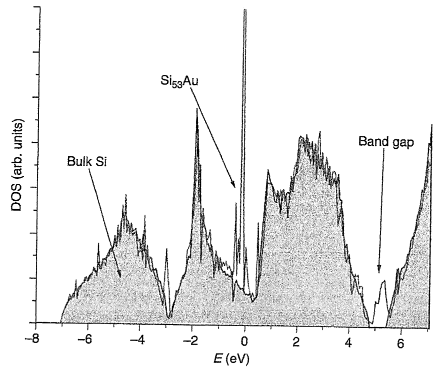

Figure 2: Electronic DOS of bulk Si (shaded region) and $Si_{53}Au$ (unshaded region).

electronic ceramic, used to make capacitors for scores of electronic devices. The favorable electronic properties of $BaTiO_3$ arise directly from hybridization (mixing) between the Ti d states, which are formally empty and sit at the bottom of the conduction band, and the O p states, which are filled and sit at the top of the valence band. This mixing should be visible in a DOS plot as an overlap between the Ti d states and the O p states (in fact, DFT calculations were used to prove the mechanism through which $BaTiO_3$ is such a useful dielectric!). Calculate the orbital-resolved DOS for $BaTiO_3$ – do you seen an overlap between these states? You will first need to optimize and converge the lattice parameter with respect to the cutoff energy and **k**-point mesh.

Exercise 2: Calculation of elastic constants

In this exercise, we will calculate the elastic constants (C_{11}, C_{12} and C_{44}) of diamond and Si. We will use the elastic constants of Si in the next homework assignment, where we will need them to calculate some thermal expansion properties of Si.

We first consider an isotropic deformation of the diamond lattice. This corresponds to a uniform stretch of the lattice vectors $\boldsymbol{a_i}$ by η:

$$\boldsymbol{a_i'} = (1+\eta)\boldsymbol{a_i} \text{ for } i = 1,2,3. \tag{1}$$

The resulting change in the total potential energy from equilibrium (undistorted) value U_0 is

$$U - U_0 = \Omega\frac{3}{2}(C_{11} + 2C_{12})\eta^2, \tag{2}$$

where Ω is, as usual, the unit cell volume. Our task is to determine C_{11} and C_{12} by calculating U and U_0. Unlike the previous exercises, instead of using the internal Espresso definitions of the lattice vectors (through the `ibrav` keyword), here we will define the lattice vectors ourselves. Set `ibrav = 0` then add an additional block of keywords under the `K_POINTS` block:

```
CELL_PARAMETERS alat
-0.5d0 0.d0 0.5d0
0.d0 0.5d0 0.5d0
-0.5d0 0.5d0 0.d0
```

The `alat` keyword specifies that we are listing the lattice vectors in units of the lattice parameter, `celldm(1)`. Run a self-consistent field calculation at the relaxed lattice parameter determined in the previous exercise at the relevant plane wave cutoff and $\boldsymbol{k}$-point mesh density. Note the total energy (your value will be slightly different):

$U_0 =$ -23.05766177 Ry.

Also take note of the unit cell volume, Ω = 74.3715 bohr^3 (again, your value will be slightly different).

Now we need to modify the lattice vectors to set an isotropic deformation with η = 0.002:

```
CELL_PARAMETERS alat
-0.501d0 0.d0 0.501d0
0.d0 0.501d0 0.501d0
-0.501d0 0.501d0 0.d0
```

The total energy of diamond with respect to this deformation is:

$U =$ -23.05761049 Ry.

Using Equation 2 with η = 0.002 we obtain $C_{11}+2C_{12}$ = 0.1149186 Ry/bohr^3 = 1690.5 GPa (1 Ry/bohr^3 = 14710.5 GPa).

We need another equation that relates C_{11} and C_{12} in order to determine their values, so we now consider a tetragonal deformation using the relation,

$$U - U_0 = \Omega 3(C_{11} - C_{12})\eta^2. \tag{3}$$

The unit cell can be deformed consistent with this distortion by considering an expansion (1+η) along x and y and a contraction (1-2η) along z. Modify the `CELL_PARAMETERS` block to take account of this deformation and repeat the procedure above to evaluate $C_{11} - C_{12}$.

Finally, we can apply a trigonal deformation to the unit cell to calculate C_{44} using the relation

$$U - U_0 = \Omega \frac{1}{2} C_{44}\eta^2. \tag{4}$$

The change in the lattice vectors due to this deformation can be seen by explicitly considering, for example, the change in $\boldsymbol{a}_1$: $a'_{1x} = a_{1x} + \eta a_{1y}/2$, $a'_{1y} = a_{1y} + \eta a_{1x}/2$, and $a'_{1z} = a_{1z}$. The other two lattice vectors change in an analogous way.

2A Calculate C_{11}, C_{12} and C_{44} for both diamond and Si following the procedure above. Calculate the total energy as a function η for η = -0.006, -0.004, -0.002, 0.002, 0.004 and 0.006. Fit the energy change per unit volume (ΔU/Ω) for each value of η to Equations 2, 3 and 4 to obtain the elastic constants (note that the zero distortion point must also be part of your fit: ΔU/Ω = 0, η = 0).

2B How do your results compare with experiment and with other calculations from the literature?

Exercise 3: Automating calculations with scripts

By now you have probably noticed that calculating certain properties with DFT involves repeating calculations a number of times, while changing the value of one parameter. Scripts can help make your calculations more efficient and reproducible. Modify the script provided in Canvas to automate the calculation of elastic constants in Exercise 2A. Make sure your script is documented such that another person could use it without having to consult you (it's fine if you want to write your own script). Submit the script along with your homework assignment write-up.

MSE 5720 Assignment Grading Rubric

Topic and points	Excellent (90 – 100%)	Good (80 – 90%)	Ok (65 – 80%)	Poor (<65%)	Points
Introductory paragraph 10 pts possible	Succinct, well written, important aspects of assignment clearly highlighted, student shows advanced understanding of subject matter.	Clear but wordy or convoluted, good understanding of assignment subject matter.	Too brief or hard to follow, superficial understanding of assignment subject matter.	Section missing or impossible to follow, essentially a regurgitation of assignment text.	
Data analysis 20 pts possible	Techniques used are relevant and appropriate to the data produced in the assignment, execution is excellent.	Techniques used are appropriate but execution is spotty.	Relevant techniques identified but significant issues with execution.	Data analysis absent or poorly executed.	
Data presentation 20 pts possible	All aspects of figures and tables clearly labeled using good graphic design principles, all axes labeled with appropriate units, scientific conventions followed.	Most figures and tables presented properly, minor issues with presentation and/or missing labels or units.	Issues with several figures and/or tables, missing/incorrect/inappropriate units, barely adequate graphic presentation.	Poor presentation of data throughout report, incomprehensible figures or tables.	
Summary 10 pts possible	Clearly synthesizes main points of assignment, challenges and alternative approaches clearly and succinctly discussed.	Some of the main points, challenges and alternative approaches identified, writing acceptable.	Included, but missing some key points and writing hinders reading.	Section missing or a restatement of assignment summary.	
Overall layout of report 10 pts possible	All sections clearly labeled, text and figures/tables arranged in a way that flows logically and helps reader understanding	Sections clearly labeled, different arrangement of text and data may improve flow and reader understanding.	Some sections clearly labeled, layout needs significant improvement.	No delineation between different sections, layout illogical and/or confusing.	

Assignment-specific problem 10 pts possible	Scripts have the proper matrix multiplication for the corresponding alat or bohr setting. Code is well documented with comments, and reads cleanly.	Scripts have the proper matrix multiplication for the corresponding alat or bohr setting, but lacks documentation/comments.	Scripts have been changed and an attempt was made to automate the process, but unsuccessfully.	Exercise missing or answer indicates lack of understanding.	

Comments: